EEG: From Signals to Insights

BioRTC 2026 - Yobe State University

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Universitat Autònoma de Barcelona

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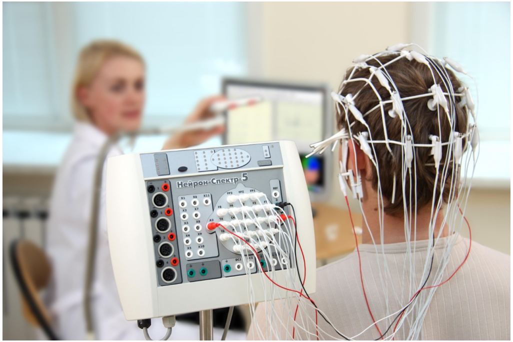
Overview

Motivation



EEG is a cornerstone of clinical and research practice, yet the physical principles behind the signal are easy to take for granted.

This lecture revisits those foundations - so that when you use EEG as a research/diagnostic tool, you know what your signals actually mean.

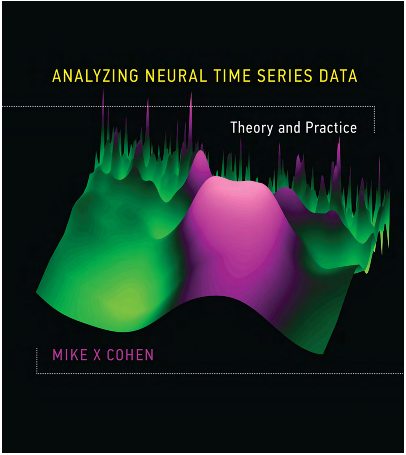
Main Reference

Most concepts, figures, and notation draw on:

Cohen, M.X. (2014). *Analyzing Neural Time Series Data: Theory and Practice*. MIT Press.

Original figures are generated from TUH EEG Corpus[a]

[a] Access by request: sign the data use agreement and email it to `help@nedcdata.org`.



The Data We Will Work With

All hands-on material uses the Temple University Hospital EEG Corpus (TUH EEG) - one of the largest open clinical EEG collections, with standardized referential recordings.

TUAR v3.0.1

Artifact annotations
(eye, muscle, electrode)

→ Preprocessing & montage demos

TUSZ v2.0.6

Seizure start/stop labels
clinical EEG, 25k sessions

→ Project 1: CNN seizure detection

TUEP v3.1.0

Epilepsy / no-epilepsy
patient-level labels

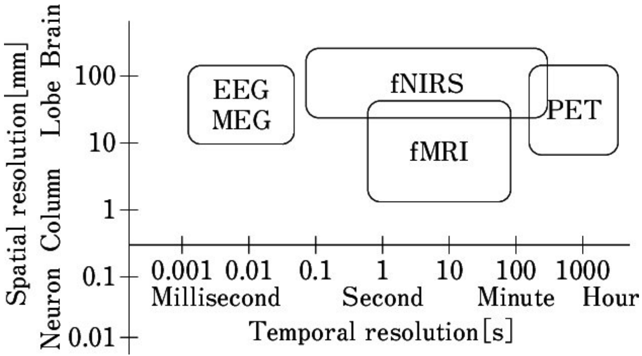
→ Project 1 extension: epileptic vs. non-epileptic classification

Every concept in this lecture is a tool you will need to work with these recordings.

Why EEG? Trade-offs & Complementarity

The Brain Imaging Landscape

- EEG: millisecond temporal resolution, low spatial resolution
- fMRI: millimetre spatial resolution, seconds lag (BOLD)
- The research/clinical question drives the method

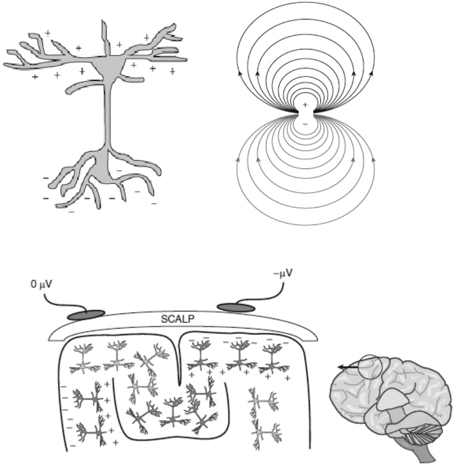


Source: Kotaro Takeda, ResearchGate

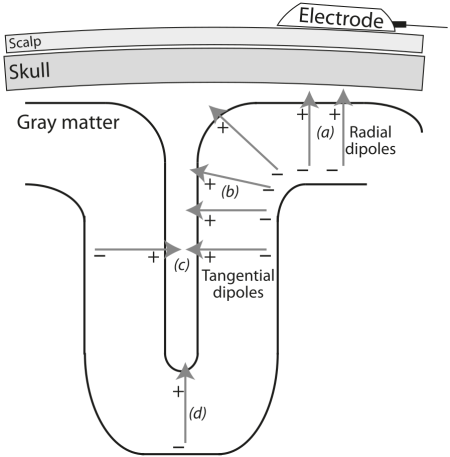
Reason 1: Direct Measurement of Neural Activity

EEG voltage fluctuations are direct biophysical reflections of synchronous postsynaptic potentials in cortical pyramidal neuron populations.

→ EEG measures the electrical source directly at the scalp



Where Does the EEG Signal Come From?



Cohen (2014), Fig. 5.1

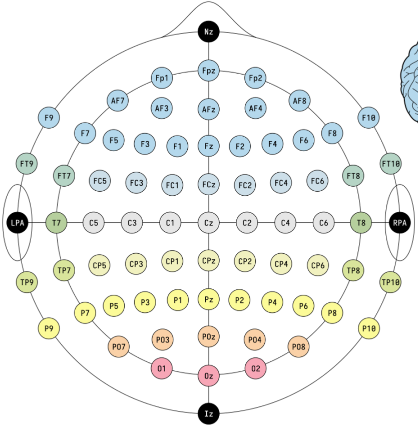
EEG measures **synchronous postsynaptic potentials** of large cortical neuron populations - modelled as current dipoles.

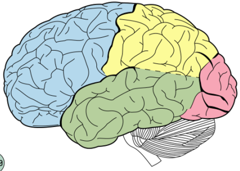
What the electrode detects depends on geometry:

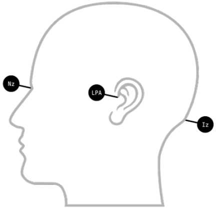
- **(a, b) Radial dipoles** on gyral crowns point toward scalp → **strongly detected**
- **(c) Tangential dipole** on sulcal wall parallel to scalp → weaker, distorted topography
- **(d) Deep source** at sulcal fundus points away → **barely detectable**

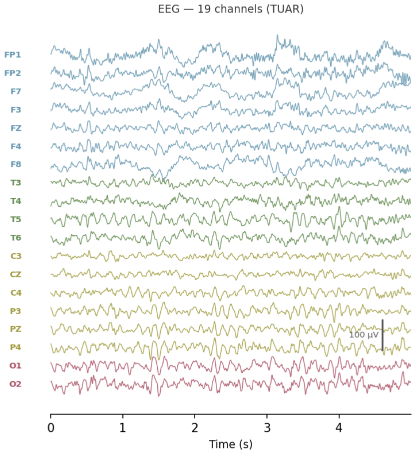
Consequence: EEG favours gyral surface activity; sulcal and deep sources are underrepresented.

EEG Measurements









Frontal Temporal Parietal Occipital

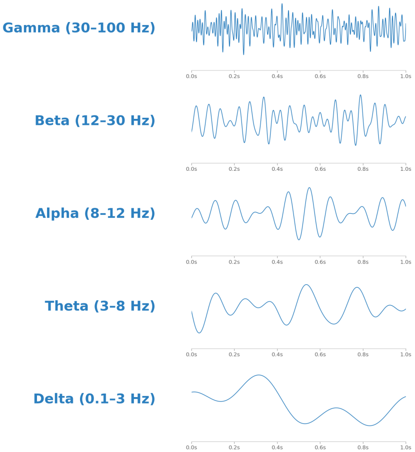
Reason 2: Temporal Resolution Matches Cognition

Cognitive processes unfold in tens to hundreds of milliseconds: brain rhythms operate at the same timescale.

Takeaway: EEG is fast enough to capture the sequential activation of the neural substrate during cognitive tasks.

Frequency bands specific to cognition

- **Gamma (30-80 Hz):** problem solving, concentration
- **Beta (12-30 Hz):** active concentration and motor control
- **Alpha (8-12 Hz):** relaxed wakefulness, sensory gating, idle state
- **Theta (3-8 Hz):** language, memory and attention, hippocampal-cortical communication
- **Delta (0.5-3 Hz):** slow-wave sleep (N3), memory consolidation



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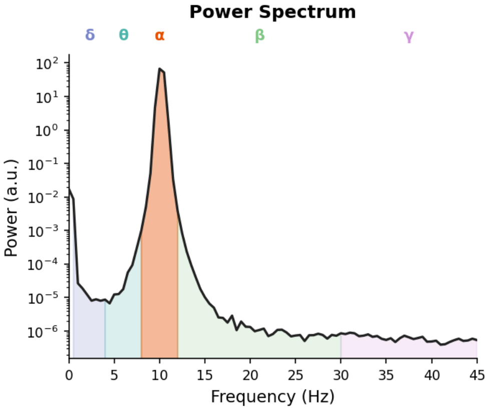
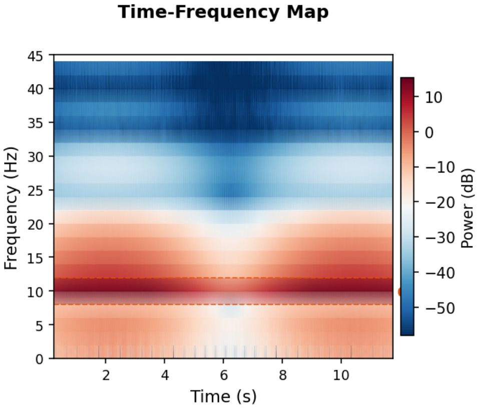
Frequency representations





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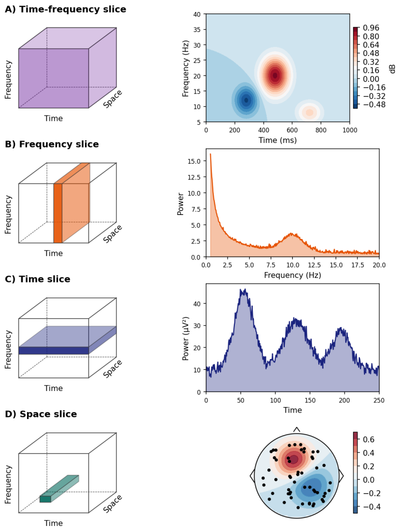
Reason 3: Multidimensional Data

EEG data span 3 dimensions: time, space (electrodes), and frequency.

Each time-frequency point yields power (signal energy) and phase (timing), giving a complete time-frequency map of brain activity.

This richness enables hypotheses grounded in both neurophysiology and psychology.



Yes, but ...

Poor spatial precision:

- Cannot resolve subcortical structures (basal ganglia, thalamus, brainstem)
- Not suited to "where in the brain" questions
- Can dissociate parietal vs. frontal - but no finer
- Deep structures largely invisible to scalp electrodes

Rule

Use the tool best suited to your research question. EEG excels at *when*; fMRI excels at *where*.

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EEG Signal Types

Oscillations, ERPs, and Induced Power

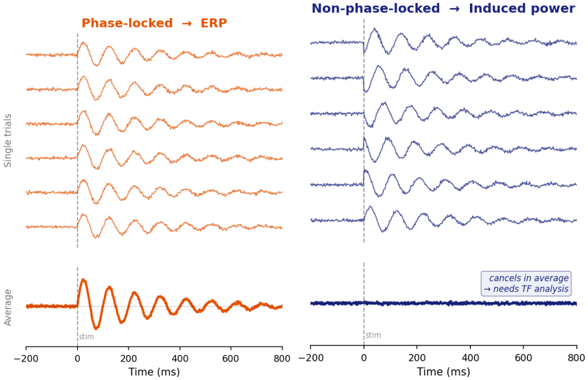
Each stimulus repetition = one **trial**. Experiments average many trials per condition.

ERP (evoked):

- Phase-locked to stimulus
- Survives trial averaging
- N1, P2, N400, P300 . . .
- frequency limited

Induced power:

- Not phase-locked
- Lost in averaging
- Requires time-frequency analysis



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ERP vs. Time-Frequency Analysis

ERP	Time-Frequency
Fast to compute, few parameters	Captures non-phase-locked dynamics
Highest temporal precision	Links to neurophysiological mechanisms
Decades of literature	Bridge across species and scales
Quick data quality check	Reveals more of the EEG signal space
Null results hard to interpret	Temporal smoothing reduces precision
Cannot test interregional synchrony or cross-frequency coupling	Many parameters $\rightarrow$ risk of "paralysis of analysis"

Both have a place. Use ERPs when timing is critical/slow frequencies; use TF when you care about oscillatory mechanisms or non-phase-locked activity.

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Preprocessing: Decisions with Consequences

Preprocessing Is Not a Checklist

Now that you know *what* you are measuring, each step has a clear motivation:

- **ERP** depends on consistent phase across trials $\rightarrow$ artifact rejection & ICA protect the average from single-trial contamination
- **Induced power** lives in specific frequency bands $\rightarrow$ filter choices determine which bands survive, (e.g. an aggressive highpass will swamp your delta analysis)
- **Spatial topographies** reflect source geometry $\rightarrow$ reference choice either preserves or washes out local gradients
- **Bipolar montage** subtracts adjacent electrode pairs $\rightarrow$ acts as a spatial high-pass filter: enhances local sources, suppresses common-mode activity, but loses absolute topography

Without this context, preprocessing feels arbitrary.

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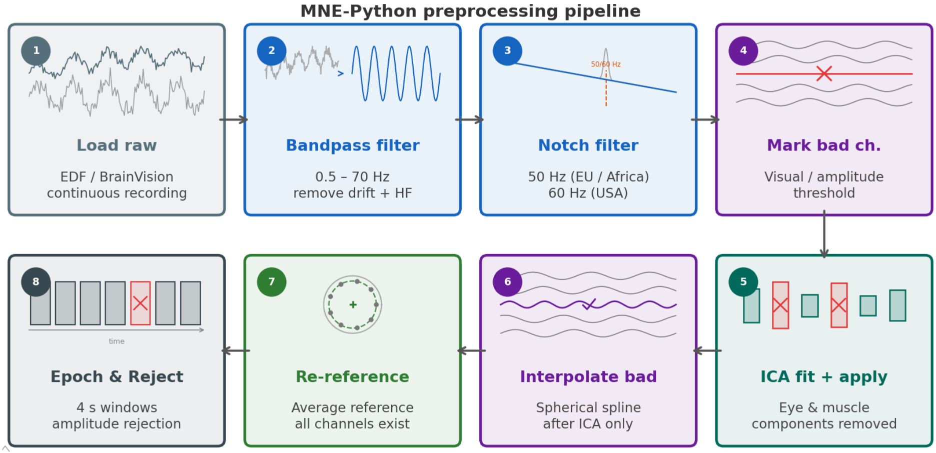
The Preprocessing Pipeline



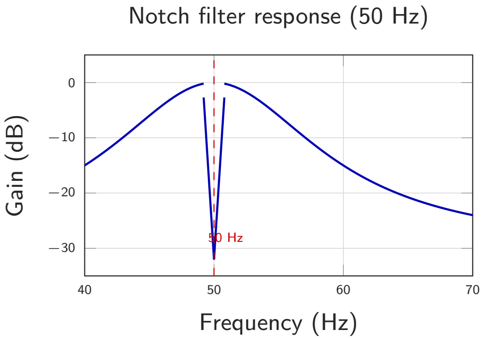
Every step is a choice. Different labs make different choices - know why you make yours.

Notch Filter: Removing Line Noise

The problem: the power grid contaminates EEG recordings through electromagnetic coupling with the recording cables.

- Appears as a sharp spike in the PSD, often with harmonics (100/120 Hz)

How it works: a *bandstop* (notch) filter suppresses a very narrow frequency band while leaving everything else intact.



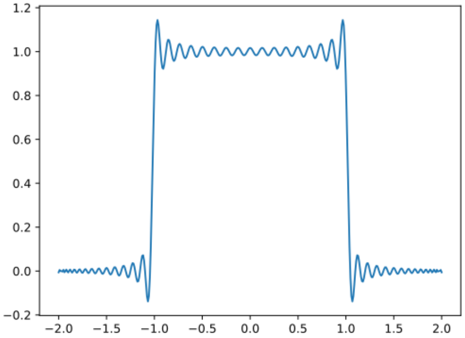
In Nigeria: use `notch_filter(50)`.

Trade-off:

- Very narrow → less signal distortion, but needs precise centre frequency
- Very wide → removes more noise, but also removes real gamma activity near 50/60 Hz

Filtering Choices

- **Highpass** (> 0.1 Hz): removes slow drift - but can distort ERPs if set too high
- **Lowpass** ($<$ 40-100 Hz): removes muscle artifact - but cuts gamma if set too low
- Higher order $\rightarrow$ sharper roll-off, but ringing spreads sharp transients in time (Gibbs)
- Rule of thumb: use moderate order + `filtfilt`; filter the full recording *before* windowing



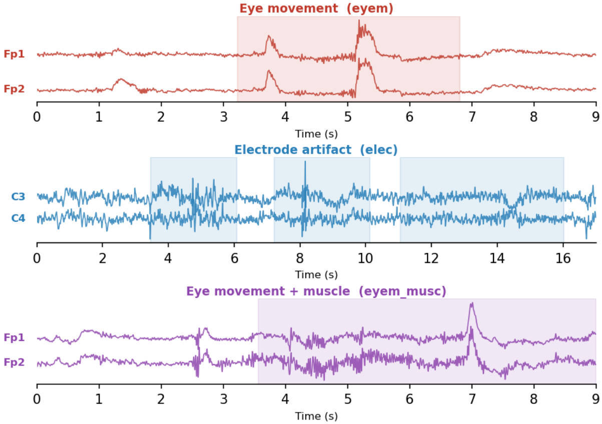
Gibbs phenomenon: ringing at sharp edges.

Source: johndcook.com

For ML/DL: ringing near seizure onsets inflates band-power features at window edges - some end-to-end models (EEGNet, ShallowConvNet) skip explicit filtering and let the network learn its own frequency decomposition.

Common EEG Artifacts

- **Ocular** (`eyem`): slow, large deflection at frontal electrodes; caused by blinks and saccades
- **Electrode** (`elec`): sudden noise or flat signal; poor contact, gel drying, or cable movement
- **Muscle** (`musc`): broadband high-frequency burst (>30 Hz); speaking, chewing, or head movement

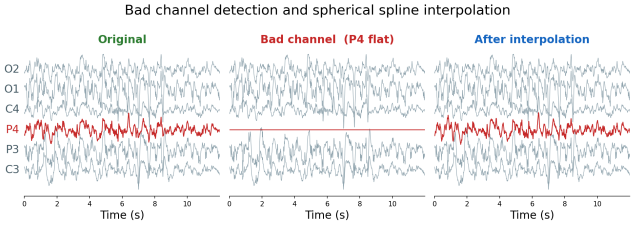


Real TUAR recording `aaaaabbn_s010_t000.edf`, annotations from companion CSV

Bad Channels and Interpolation

What makes a channel bad?

- **Flat line:** disconnected electrode or broken lead
- **Constant high noise:** poor skin contact, high impedance
- **Slow drift:** electrode movement or sweat
- **Bridging:** two electrodes short-circuited together



P4 zeroed (disconnected lead) $\longrightarrow$ MNE spherical spline interpolation; interictal segment, TUSZ aaaaasgd

Spherical spline interpolation:

Replace the bad channel with a smooth weighted average of its neighbours, respecting the geometry of the scalp.

Order matters: flag bad channels *before* ICA - one bad channel spreads into many components and biases the decomposition.

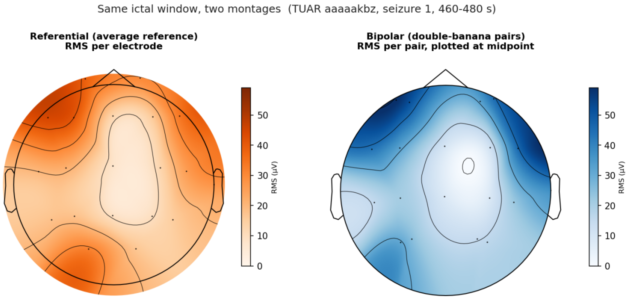
Referencing: A Real Methodological Choice

Referential (monopolar):

- Each electrode vs. a common reference
- Flexible: re-reference, ICA, source reconstruction
- Sensitive to reference quality

Bipolar (difference):

- Subtract adjacent electrodes
- Cancels common-mode noise
- Standard for clinical EEG reading
- Less flexible - cannot easily undo

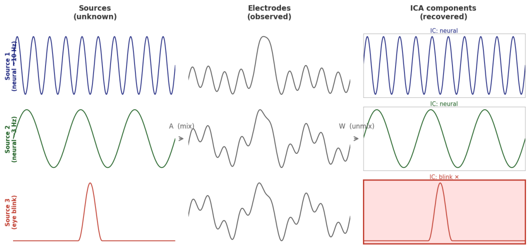


RMS amplitude, same 20 s ictal window (TUAR aaaaakbz, seizure 1). Referential (left) has a real value at all 19 electrodes; bipolar (right) only at each pair's midpoint - a sparser, uneven grid that smooths the map for geometric reasons alone, independent of any true signal difference. Even so, both still agree on the same fronto-temporal hotspot (F7 / Fp1-F7).

ICA: Independent Source Separation

Cocktail party: many voices, many microphones. Weighted combinations of microphone signals can isolate one voice, ICA finds those weights.

In EEG: electrodes = microphones, sources = voices.



Electrodes observe *mixtures* (A); ICA inverts the mixing matrix (W).

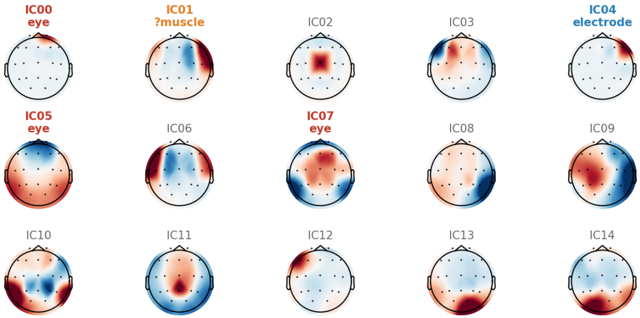
Why not PCA?

- PCA: orthogonal axes of max *variance* - loudest combinations, not separable sources
- ICA: statistically independent components - uses higher-order statistics (non-Gaussianity)
- Artifacts are sparse, non-Gaussian - a blink is a rare spike, not Gaussian noise
- PCA spreads one blink across many components; ICA isolates it in one

ICA for artifact identification

ICA decomposes EEG into independent spatial components - each with a characteristic topography:

- **Eye** (IC00, IC05, IC07): frontal; Fp1/Fp2 dominant
- **Electrode** (IC04): focal single-channel; F8 only
- **Muscle** (IC01): right temporal (F8/T8) - *ambiguous*: could also be neural



■ eye ■ electrode ■ ?muscle (ambiguous) - removed: eye + electrode

TUAR aaaaabbn_s010_t000, average reference, 19 channels, ICA n=15, 1-40 Hz

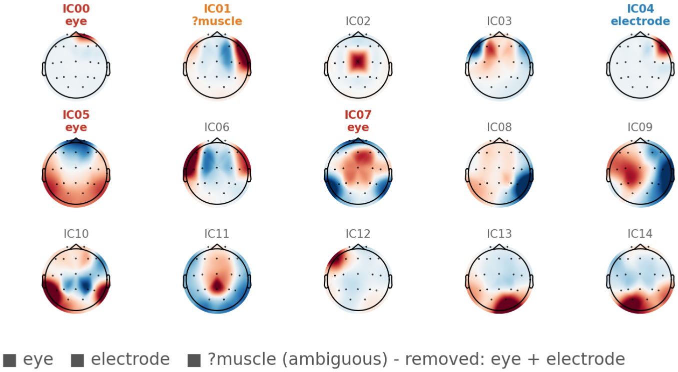
Filter first, then ICA: a 30 Hz lowpass removes most muscle before ICA runs - IC01 would largely vanish. ICA handles what filtering cannot: slow ocular and focal electrode artifacts.

Pitfall: removing ambiguous components (IC01) risks discarding real neural signal.

ICA Labeling: Beyond Eyeballing the Topography

Colour alone can mislead. Two quantitative checks catch what eyeballing misses:

- **Channel-weight isolation:** is one electrode's weight far larger than every other?
- **Time-course kurtosis:** a quiet baseline with occasional huge spikes → bad contact, not brain signal

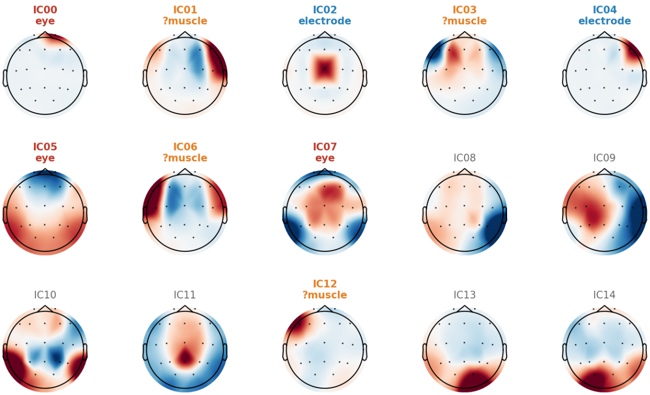


TUAR aaaaabbn_s010_t000, average reference, 19 channels, ICA n=15, 1-40 Hz

ICA Labeling: Beyond Eyeballing the Topography

Applying both to the *same* 15 components finds 4 more worth a second look:

- **IC02**: Cz-isolated (1.35 vs. 0.17), kurtosis 87 - same signature as IC04 → another **electrode** artifact
- **IC03, IC06, IC12**: lateral-frontal, beta-heavy, kurtosis 24-41 - same profile as IC01 → **ambiguous**, not confidently neural



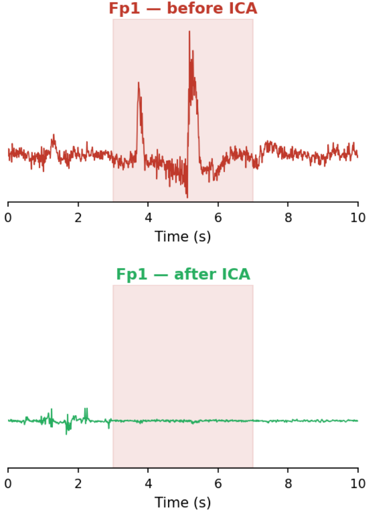
■ eye ■ electrode ■ ?muscle (ambiguous) - - - newly identified this pass

Identified by checking channel-weight isolation and time-course kurtosis, not topography colour alone.

TUAR aaaaabbn_s010_t000, average reference, 19 channels, ICA n=15, 1-40 Hz

Payoff: A Cleaner Signal

Takeaway: "doesn't look like an obvious artifact" is not the same as "confirmed clean." Always check the time course and channel weights, not just the topography colour.



The 1/f Problem

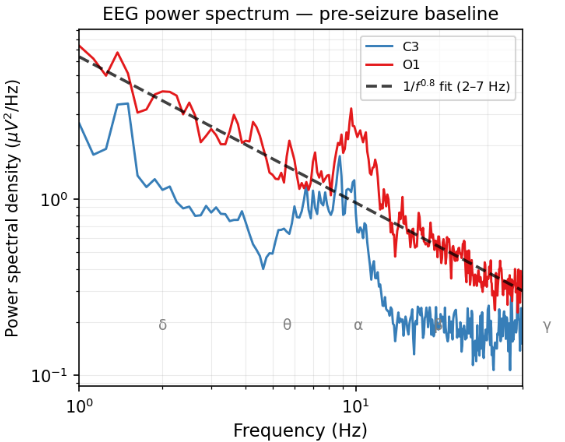
Neural signals have power that falls with frequency:

$$P(f) \propto \frac{1}{f^{\alpha}}, \quad \alpha \approx 1\text{-}2$$

This is universal in EEG - not a pathological feature.

Consequence: low frequencies always dominate any power map.

Solution: divide each frequency by its baseline power $\rightarrow$ *baseline normalization*.



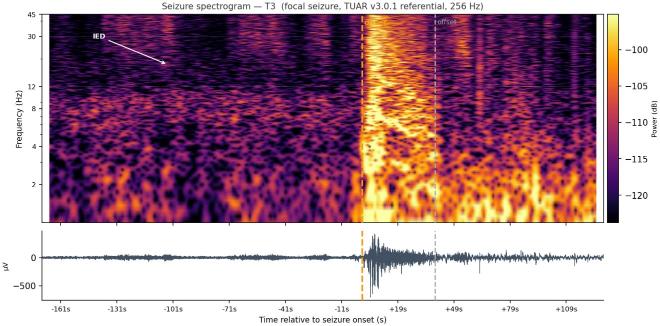
Power spectral density (Welch). Channels C3 and O1, pre-seizure baseline.

Note the alpha peak (~10 Hz) in O1: this patient had eyes closed.

1/f in Time-Frequency representation

The same 1/f slope dominates a time-frequency map.

Low frequencies (δ, θ) saturate the colorscale - high frequencies (β, γ) are invisible even when they carry seizure information.



Absolute log power. Channel T3 (left temporal). Focal seizure onset $t = 0$.

Removing 1/f: Baseline Normalization

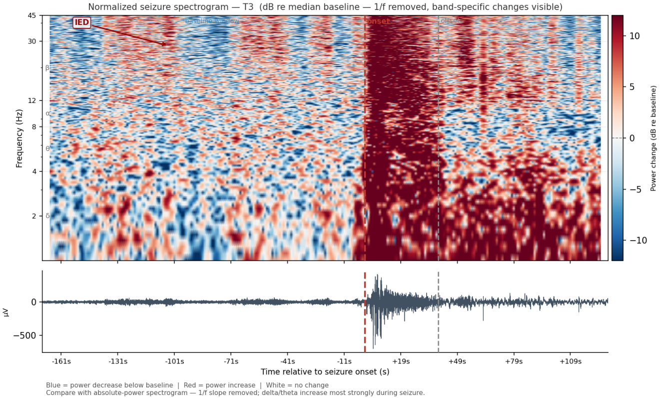
Divide by **median power in a pre-seizure window** per frequency:

$$P_{\text{norm}} = 10\log_{10}\left(\frac{P(f,t)}{P_{\text{bl}}(f)}\right)$$

- 1/f removed - all bands comparable
- δ/θ increase most during seizure
- White = baseline; red = increase; blue = suppression

Why IEDs show in β/γ only?

- Low f: large δ/θ background *masks* IED contributions.
- High f: quiet, steady baseline - brief broadband burst *stands out*.

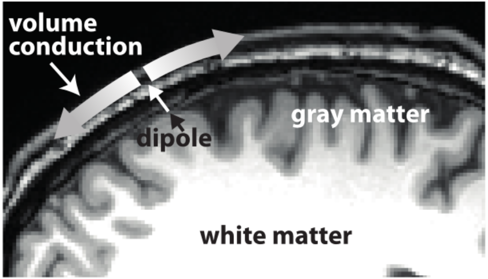


dB re median baseline. Same recording and channel.

IED: interictal epileptiform discharges.

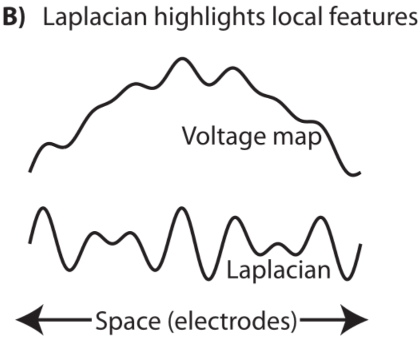
Volume Conduction: Why One Source Looks Like Many

A) Depiction of volume conduction



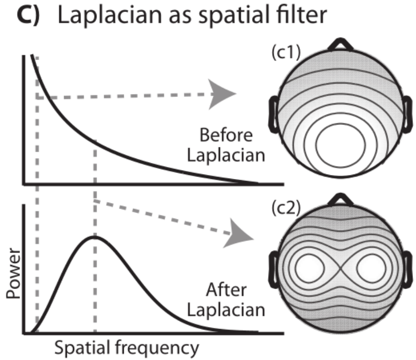
A dipole's field spreads passively through tissue - no electrode sees *"its own"* source alone.

Volume Conduction: Why One Source Looks Like Many



The raw voltage map blurs sharp, local features.

Volume Conduction: Why One Source Looks Like Many



Laplacian = spatial high-pass filter: boosts local content, suppresses the shared blur.

Why this matters: volume conduction is why one artifact source spreads across many electrodes - spatial filters (Laplacian/CAR) can sharpen what volume conduction blurs.

Adapted from Cohen (2014), Fig. 22.1.

Spatial Filtering: One Idea, Many Names

A spatial filter is a weighted combination of channels:

$$y(t) = \mathbf{w}^\top \mathbf{x}(t)$$

Different weight vectors **w** target different goals:

- **CAR** / **Laplacian** - suppress common noise, sharpen local topography
- **ICA** - unmix sources by statistical independence
- **CSP** (common spatial patterns) - choose **w** to maximize the variance ratio between two classes (e.g. seizure vs. background)
- **Beamforming** - one **w** per brain location, later in this lecture

Same equation throughout the course - only how **w** is chosen changes.

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Epoching, Windowing, and Rejection

Epoching: cut continuous data into event-locked segments:

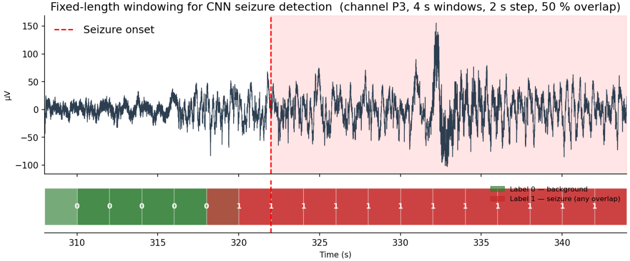
- Typical: -200 to $+800$ ms around stimulus
- Baseline correction from pre-stimulus interval
- Each epoch is independent by design

Rejection: after ICA, flag residual artifacts:

- Amplitude threshold ($\pm 100\,\mu$V)
- Joint probability / kurtosis
- Visual inspection

We will not apply rejection in our CNN

Windowing (our CNN pipeline)



TUSZ aaaaasgd - 4 s windows, 50 % overlap; red = any overlap with seizure onset

Label strategy matters: “any overlap = seizure” vs. “>50 %” - same data, different class balance and model sensitivity.

Artifacts Are Not Just Noise - They Are Features

Unpreprocessed EEG fed to a CNN does not just add random noise. Artifacts are systematic - the model will learn them as classification features:

Muscle artifact → false seizures

EMG broadband noise (>30 Hz) mimics high-frequency oscillations that appear in real seizures. A CNN learns: "high-freq burst at T7/T8 = seizure" - correct during training, fires on *chewing, movement, electrode bumps* in deployment.

Label contamination

A 1-s window spanning an eye blink is labelled "non-seizure" but looks anomalous - the model trains on a noisy label.

Domain shift: TUH → Yobe

TUH: US hospital, 60 Hz line noise, specific gel.

Yobe: 50 Hz, different equipment, different patient movement profiles.

A model that learned TUH artifact signatures will fail to generalize even if seizure morphology is the same.

Hypothesis to test in the workshop

Preprocessing before training helps the model learn *seizures*, not equipment signatures - but does it *always* help?

Time-Frequency Analysis: A Decision Tree

Four Approaches - When to Use Which?

Welch PSD
Averaged FFT
For stationary segments
Stable power estimate
No time resolution
used for band power

STFT
Fixed window
Fast, interpretable
Long window suppresses brief transients
used in our spectrogram

Morlet Wavelet
Adaptive window
Best for neural data
Intuitive parameters (f_0, cycles)

Hilbert Transform
Narrow-band filtered
Fast, good for single bands
Requires careful filtering

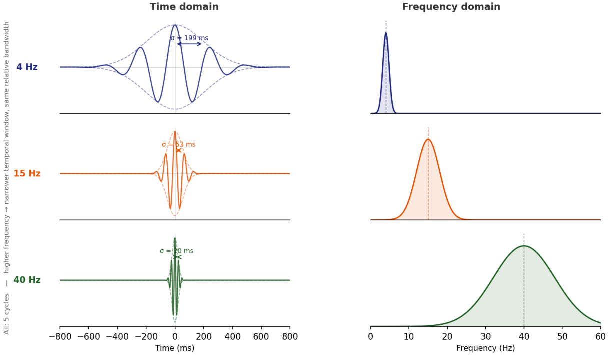
Heisenberg Uncertainty

You cannot have perfect time *and* frequency resolution simultaneously. Wavelet cycles control this trade-off.

Morlet Wavelets: Intuition

- Gaussian-windowed sine wave at frequency f_0
- **Number of cycles** n: more cycles $\rightarrow$ better frequency resolution, worse temporal precision
- Typical range: 3-10 cycles; scale with frequency
- Convolution via FFT $\rightarrow$ efficient

Same 5 cycles at all three frequencies - same *relative* bandwidth, but temporal window shrinks at higher f_0.



Connectivity: ISPC and Coherence

Connectivity: Synchrony, Coherence, and Volume Conduction

Why connectivity?

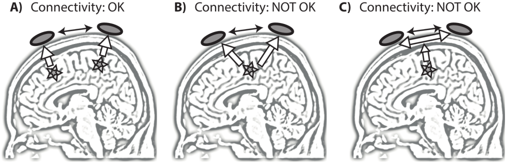
- The brain is a network - local power is not enough
- **ISPC/PLV** (Inter-Site Phase Clustering/Phase locking value): phase synchrony between channels across trials; frequency-specific
- **Coherence:** phase + amplitude consistency; normalized 0-1; can be driven by amplitude co-variation alone
- **Effective connectivity** (Granger, DCM): adds directionality - beyond today's scope

Volume conduction - the main confound

One source → many electrodes simultaneously → spurious synchrony at zero lag.

A) Two distinct sources → genuine connectivity
B) One source, two electrodes → spurious synchrony
C) Common reference → reference artefact

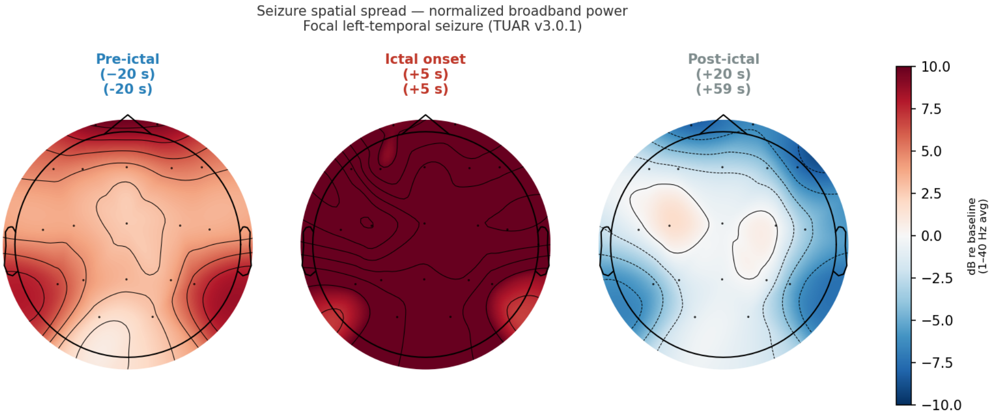
Mitigations: surface Laplacian, imaginary coherence, source-space connectivity



Silva Pereira et al., *Brain Connectivity*, 2017

Machine Learning for EEG: Seizure Detection

Seizure Spatial Spread



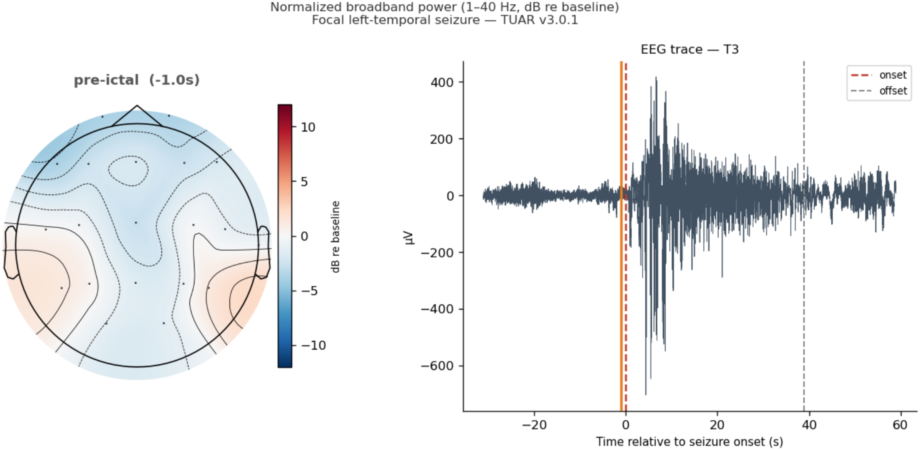
Quiet baseline - normal EEG patterns

Rapid generalization from left temporal focus

Post-ictal suppression - whole-brain power drop

Normalized broadband power (1-40 Hz, dB re baseline). Same recording as spectrograms.

Seizure Spatial Spread - Animation



Normalized broadband power (1-40 Hz). 2× real time.

Seizures Are Not One Disease

Focal onset

Starts in one network; may stay local or spread.

- **fnsz** - focal, non-specific
- **cpsz** - impaired awareness ("complex partial", older term)

Generalized onset

Engages bilateral networks from the start.

- **absz** - absence (brief, non-motor)
- **gnsz** - generalized, non-specific
- **tcsz** - tonic-clonic (convulsive)
- **tnsz** - tonic

Based on the ILAE 2017 operational classification. TUSZ pools **all** of these under one binary label (`seiz` vs. `bckg`) - your classifier must generalize across very different generators, durations, and scalp signatures.

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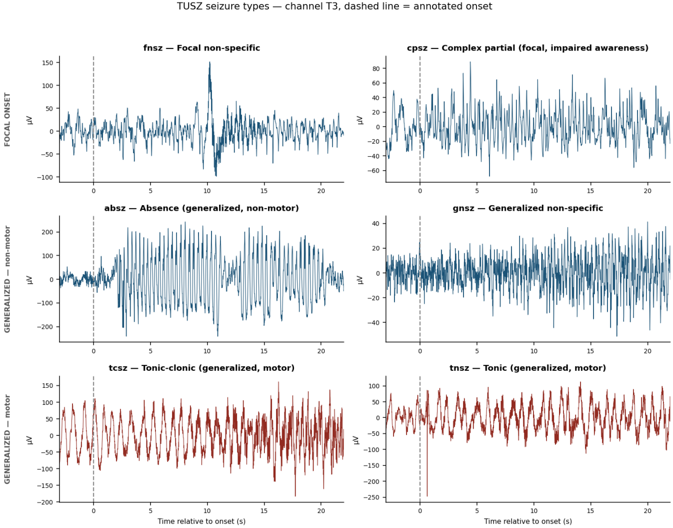
What They Look Like

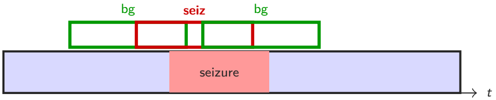


Channel T3, 25 s window around annotated onset (dashed line). Six TUSZ patients, six seizure types - same electrode, very different morphology and amplitude.

From Signal to Label: Windowed Classification

The windowing step

- Slide a 4-second window across the recording (2s hop)
- Each window $\rightarrow$ one training example
- Label: seizure if $\geq$50% of window overlaps an annotated ictal interval; background otherwise



Each window gets one label; the model never sees adjacent context - it must decide from a fixed 4s snapshot.

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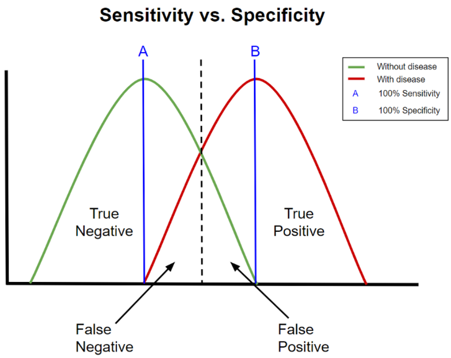
Sensitivity and Specificity: The Clinical Trade-off



Sensitivity (recall, TPR)

Proportion of seizures correctly detected

$$\text{Sens} = \frac{TP}{TP + FN}$$

Specificity (TNR)

Proportion of background correctly rejected

$$\text{Spec} = \frac{TN}{TN + FP}$$

Threshold trade-off

Moving threshold **left** $\Rightarrow \uparrow$ sensitivity, $\downarrow$ specificity.

- maximize sensitivity - missed seizure can be fatal
- balance - alarm fatigue leads to ignoring alerts

The Class Imbalance Problem and the Accuracy Trap

Typical seizure recording

- 30 min recording, 76 s total seizure
- Background : Seizure $\approx$ 22 : 1
- A model that always predicts "background" achieves **96% accuracy**
- Sensitivity = **0%** - useless clinically

Lesson

Accuracy is a **misleading metric** under class imbalance.
Report: **sensitivity** (seizure recall) + **specificity** (background recall)

Fixes for imbalance

- **Weighted sampler**: oversample seizure windows in each mini-batch
- **Class weights**: upweight minority class in cross-entropy loss
- **Threshold tuning**: lower decision threshold $\rightarrow$ more sensitive, less specific (ROC curve)

Clinical tradeoff: missed seizure (false negative) vs. alarm fatigue (false positive) - context determines which matters more.

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Time Split vs. Random Split: The Leakage Trap

Random split ← common mistake

- Windows from the *same seizure* land in both train and test
- Model "recognizes" seizure it has already seen fragments of
- Reported sensitivity: optimistic & non-generalizable

Time split ← realistic

- Train on early recording; test on late recording
- No data leakage: test seizure is genuinely unseen
- Enables **online metric**: detection *latency*

Online evaluation

After the time cut, slide the model forward in time.

Latency = time from seizure onset to first correct detection.

Clinical target: <10 s for bedside alert; <30 s for automated response.

In the workshop: train on seizures 1+2 (322 s, 1042 s); test on seizure 3 (1467 s). How quickly does the model fire?

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The Generalization Ladder

Step 1 random split, same recording

Have: labels, same morphology. Risk: optimistic result - model has seen similar windows during training.

Step 2 time split $\rightarrow$ new patient, TUH

Have: ground truth, same equipment. Risk: patient variability (seizure type, onset zone, morphology).

Step 3 Yobe: no labels, different hardware

Have: raw signal + clinical report. Risk: everything else - no annotations, different hardware, 50 vs 60 Hz.

Step 3 shifts the question from *"what is my accuracy?"* to *"what can I trust - and how do I know?"*

Source Reconstruction

Topographical vs. Brain Localization

Topographical localization

Identifying which electrodes show the effect - e.g. "the error-related negativity peaks at Cz/FCz."

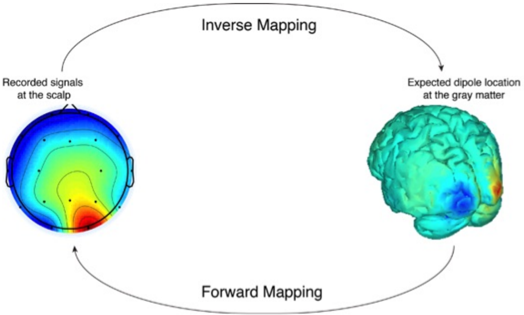
A description of what was observed. Does not by itself claim the generator sits directly under that electrode.

Brain localization

Identifying which brain regions generated the scalp activity.

An interpretation, not a direct observation - it rests on assumptions about head geometry and conductivity, and the inverse problem has many equally plausible solutions for the same topography.

Source Reconstruction: From Scalp to Brain

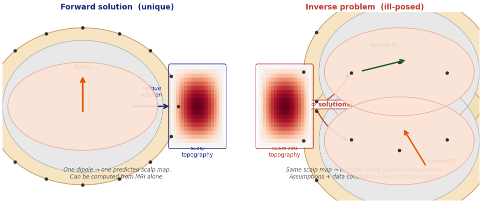


Source: Neuroelectrics

The goal: map scalp-recorded potentials onto brain source locations.

Source Reconstruction: From Scalp to Brain

- Why? Sensor-space analysis mixes signals from distant regions; volume conduction inflates apparent connectivity
- What you get: an activation time course (or TF map) for each brain voxel or cortical patch
- What you need: head model (MRI or sphere), source space, inverse algorithm



Forward: unique. Inverse: ill-posed.

"Electrodes are not the brain. If you want to make claims about brain regions, you need source imaging." - Cohen, Ch. 24

Forward and Inverse Problems

Forward (unique):

$$\mathbf{d} = \mathbf{L}\,\mathbf{s}$$

- $\mathbf{L}$: lead field matrix (encodes head geometry)
- One source config $\rightarrow$ one scalp map
- Head model: sphere (fast) or BEM/FEM from MRI

Inverse (ill-posed):

Many $\hat{\mathbf{s}}$ fit the same $\mathbf{d}$ — must add a prior.

$$\hat{\mathbf{s}} = \mathbf{L}^\top(\mathbf{L}\mathbf{L}^\top + \lambda\mathbf{I})^{-1}\mathbf{d}$$

MNE: prefer weak, widespread activity.
LORETA/sLORETA: add spatial smoothness.

λ: regularization tuned to SNR.

Resolution limit

EEG: $\sim$1-2 cm at best (skull smears fields).

Dipole Fitting and Beamforming

Dipole fitting (parametric)

Assume 1-few equivalent current dipoles. Nonlinear optimization: find (x, y, z, θ, ϕ) minimizing $\|\mathbf{d} - \mathbf{Ls}\|^2$.

- Good for: single localized generator (auditory N1, visual P1)
- Limitation: number of dipoles fixed *a priori*; fails for distributed activity

Different family: MNE/LORETA are *distributed* - thousands of fixed sources, only **magnitudes** estimated. Weights are *nonadaptive* (from the head model alone), so they work on a **single time point**.

Beamforming (adaptive spatial filter)

Weights derived from *data covariance* **R** - adapts to the actual recording.

- Passes signal from location **r**, suppresses interference from elsewhere
- Adapts to actual data - unlike fixed MNE/LORETA weights
- Needs a **time window** to estimate **R** (not one time point)
- Excellent for oscillatory activity (alpha, gamma)
- Caveat: breaks when sources are **correlated** (suppresses both)

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Source Reconstruction in Practice

With 19 channels:

- Informative but blurry (~1-3 cm error)
- Use as hypothesis, not ground truth
- Template MNI brain if no individual MRI
- Average reference before computing covariance

  (approximates reference at ∞; required for source recon)

Yobe project relevance

- Seizure onset zone localization - guides resection hypothesis
- No individual MRI in Yobe $\rightarrow$ template brain essential
- CNN detects seizures; source imaging adds *where* they start
- Week 2 focus: get preprocessing right; source imaging is the longer-term goal

"Source imaging with EEG is not perfect, but it is unique - no other method combines centimetre spatial resolution with millisecond temporal precision." - Cohen, Ch. 24

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What You Will Do

Today - Workshop
(Fri Jun 19, 1.5h)

- Preprocessing pipeline (MNE-Python)
- Time-frequency, band power
- MLP + CNN seizure detection

Hospital visit
(Tue Jun 23)

- Visit to Yobe Specialist Hospital
- Observe clinical EEG setup
- Meet the neurological team

Week 2 - Projects
(afternoons)

- **Project 1** - DL seizure detection: beat the baseline, transfer to Yobe EEG
- **Extension** - epileptic vs. non-epileptic patient classification (TUEP)

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Take-Home Messages

1. EEG's strength is time - use it; its weakness is space - be honest about it
2. Every preprocessing step is a choice - know the consequences; it can *hurt* if the artifact correlates with your target
3. Time-frequency analysis separates evoked from induced dynamics
4. EEG is a window into network dynamics - connectivity matters
5. Machine learning on EEG: accuracy is misleading; report sensitivity + specificity; time split, not random split
6. A model that works on your training data may fail silently on new data - generalization must be tested, not assumed; reporting failure is also a result

Questions?

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Supporting Slides

Feature Representations: MLP vs CNN

Two ways to encode a 4-second EEG window as model input:

- **MLP**: band-power per channel $\rightarrow (n_{win},\ n_{ch} \times n_{bands})$ matrix of scalars; discards time structure within the window
- **CNN**: raw signal $\rightarrow (n_{win},\ n_{ch},\ n_{samples})$ tensor; temporal order is preserved and exploited by the convolutional filters

Choice depends on whether fine-grained morphology (spike shape, ictal pattern) matters for your task. MLP is faster and more interpretable; CNN can learn morphological features automatically.

The Forward Problem: Physics $\to$ Scalp

Given a source configuration $\mathbf{s}$, the scalp potential is *uniquely* predicted by Maxwell's equations:

$$\mathbf{d} = \mathbf{L}\,\mathbf{s}$$

- $\mathbf{d} \in \mathbb{R}^{n_{\text{ch}}}$: electrode potentials
- $\mathbf{L} \in \mathbb{R}^{n_{\text{ch}} \times n_{\text{src}}}$: **lead field matrix** (encodes head geometry)
- $\mathbf{s} \in \mathbb{R}^{n_{\text{src}}}$: source amplitudes

Head model

Sphere: fast, approximate

BEM/FEM: from individual MRI — accurate but requires segmentation

Forward is unique

One source configuration $\to$ one predicted scalp map.

Think of $\mathbf{L}$ as the *design matrix* in a regression: computed once from anatomy; same for every time point.

With 19 channels: $\mathbf{L}$ is $19 \times \sim 10^4$ — far fewer equations than unknowns. The inverse is the hard part.

The Inverse Problem: Infinitely Many Solutions

We want $\hat{\mathbf{s}}$ given $\mathbf{d}$ and $\mathbf{L}$. The system is underdetermined: we must add a prior (regularization).

Minimum-norm estimate (MNE):

$$\hat{\mathbf{s}} = \mathbf{L}^\top(\mathbf{L}\mathbf{L}^\top + \lambda\mathbf{I})^{-1}\mathbf{d}$$

Minimize $\|\mathbf{s}\|^2$ — prefer weak, widespread activity.

LORETA:

$$\hat{\mathbf{s}} = \arg\min \|\mathbf{d} - \mathbf{L}\mathbf{s}\|^2 + \lambda\|\nabla^2\mathbf{s}\|^2$$

Adds spatial smoothness via Laplacian — more anatomically plausible.

Cohen's analogy: $\mathbf{L}$ is the predictor matrix, $\mathbf{s}$ the regression coefficients — minimize prediction error plus a penalty term (ridge / Tikhonov regularization).

Spatial resolution limits

EEG source reconstruction: ~1–2 cm at best. fMRI: ~3 mm.

More electrodes → better resolution, but there is a fundamental physical limit: skull smears the fields.

Regularization parameter λ

Too small: noisy, overfits sensor noise. Too large: over-smooth, loses focal activity. Typically set by SNR estimate or cross-validation.

Is MEG Better Than EEG for Localization?

In simulation: EEG can match or beat MEG.

- EEG is sensitive to **radial and tangential** dipoles
- MEG is sensitive mainly to **tangential** dipoles

In practice: MEG usually wins.

- EEG accuracy needs many electrodes, *measured* electrode positions, and accurate skull/scalp conductivities
- These are rarely obtained for routine EEG

Take-home

The EEG–MEG gap is mostly **practical**, not fundamental. As electrode technology and per-subject head modeling improve, EEG source reconstruction will improve.

Recall the gyral/sulcal dipole slide: the radial vs. tangential distinction is exactly why EEG and MEG "see" different sources.